Watching a flowsheet solve fail

The difflow flowsheet debugger, step by step

```
difflow debug plant.py
```

WHY A DEBUGGER

A failed solve tells you *that*, not *where*

- A recycle that will not close usually fails somewhere specific: a unit returns a NaN on the first pass, one tear field oscillates, a reactor's own solve gives up.
- `fs.solve()` warns, and `difflow.diagnostics` reads the residual history. Both describe the **end** of the solve.
- The debugger lets you **watch** the solve: it pauses unit by unit and tear iteration by tear iteration, shows every stream, and stops on breakpoints.

It is not `pdb`: `pdb` stops on lines of Python. This stops on the flowsheet's own structure, such as *before unit R1*, *at iteration 12*, *when the residual rises* or *where the first NaN is born*, and its commands speak streams, units and tears.

THE EXAMPLE

An isomerization loop with a recycle




```
# plant.py
fs = Flowsheet(species_order=species)
fs.add_feed("fresh", make_stream({...}, T=340.0, P=101325.0))
fs.add_unit(Unit("mixer", Mixer(species, thermo),
                 ["fresh", "recycle"], ["reactor_feed"]))
fs.add_unit(Unit("reactor", reactor, ["reactor_feed"],
                 ["effluent"], params={"T_spec": 340.0}))
fs.add_unit(Unit("split", Splitter(species), ["effluent"],
                 ["recycle", "product"], params={"split_frac": 0.8}))
fs.add_recycle("recycle", "recycle")    # the tear stream
```

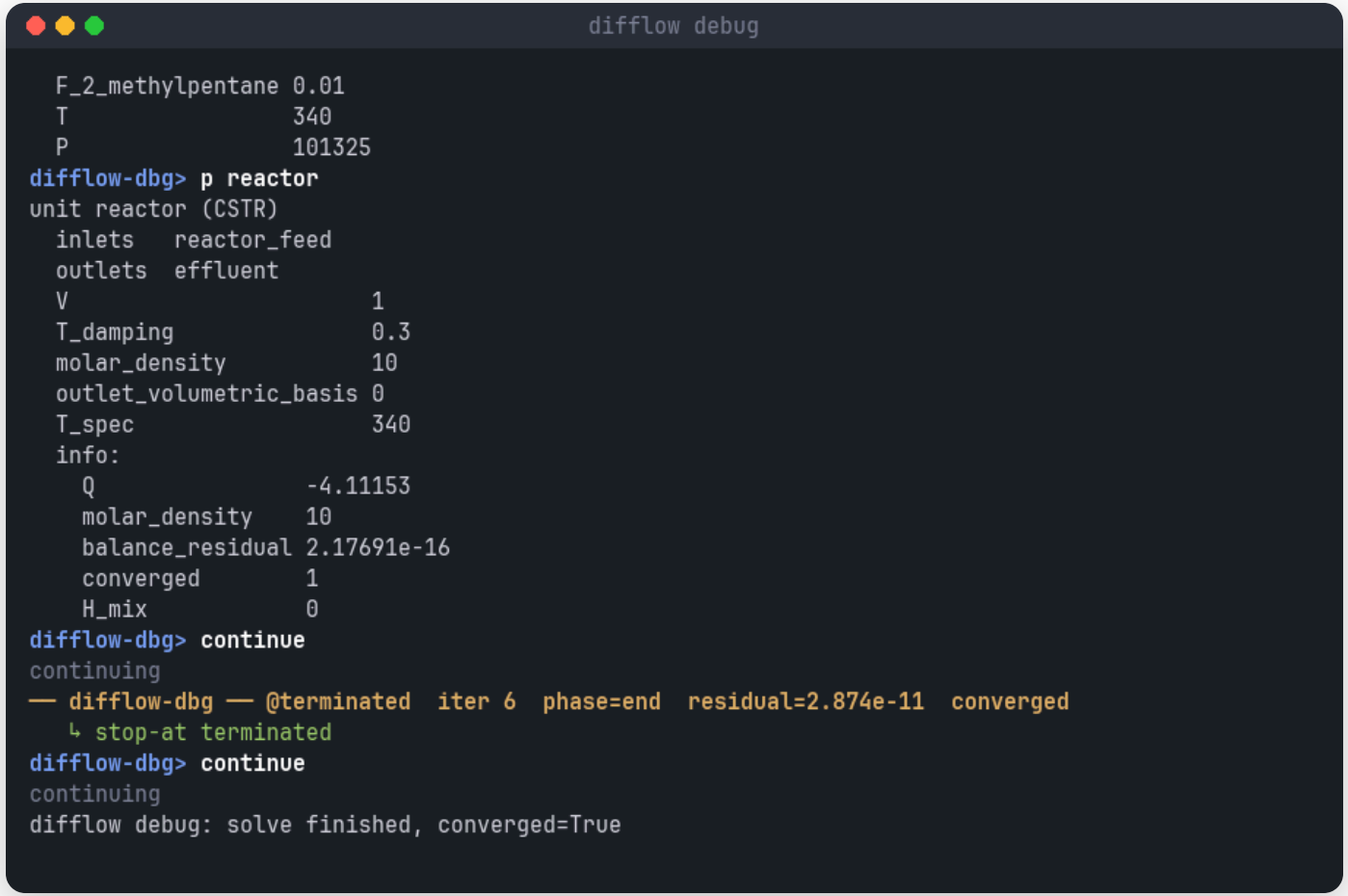
n-hexane → 2-methylpentane in a CSTR; 80 % of the effluent goes back to the mixer.

`recycle` is the **tear**: the solver guesses it, runs the units, compares what comes back, and repeats until the two agree.

All the recordings that follow are real runs of this file.

STEP 1 · START IT

`difflow debug plant.py`



It stops **before the first pass** (`solve_start`).

- `where`: the calculation order; `*` marks the tear
- `next`: run to the next unit
- `p STREAM`: a stream as it is right now
- `p UNIT`: its parameters, and what it reported (`info`)
- `continue`: run to the end (it stops once at `terminated` to show the verdict)

The first units run in the **init** phase: one pass from the feeds to guess the tear.

WHERE IT STOPS

Checkpoints and phases

Checkpoint	Fires	Fresh at that point
`solve_start`	once, before anything runs	the feeds
`before_unit` / `after_unit`	around every unit call	the inlets; then the outlets and the unit's `info`
`unit_error`	a unit raised	the inlets that broke it
`iteration`	the tear map was evaluated	`x`, `g(x)`, the residual, the history
`after_accel`	Wegstein/Anderson made the next guess	the guess (`p next`)
`terminated`	the solve is about to return	the verdict

Unit pauses carry a **phase**: `init` (the tear-guessing pass) · `solve` · `probe` (error probe after convergence) · `final`. `step` runs to the next iteration, `next` to the next unit, `stepi` to the next checkpoint of any kind.

`step`, `p tear`, `p history`

```
difflow debug
stream       field                        x          g(x)       |g-x|
recycle      F_n_hexane            0.542211      0.967411     0.4252
recycle      F_2_methylpentane     0.273789      0.485389     0.2116
recycle      T                          340           340          0
recycle      P                       101325        101325          0
difflow-dbg> step
stepping to the next iteration
— difflow-dbg — @iteration  iter 1  phase=solve  residual=3.400e-01
   ↳ step
difflow-dbg> step
stepping to the next iteration
— difflow-dbg — @iteration  iter 2  phase=solve  residual=6.525e-04
   ↳ step
difflow-dbg> p history
   0  0.4252
   1  0.340003
   2  0.000652472
reads as: falling, about x0.039 per iteration
difflow-dbg> continue
continuing
— difflow-dbg — @terminated  iter 6  phase=end  residual=2.874e-11  converged
   ↳ stop-at terminated
difflow-dbg> continue
continuing
difflow debug: solve finished, converged=True
```

`step` runs a whole pass and stops at the next **tear iteration**.

`p tear` breaks the residual down by stream and field, largest first:
`x` is the guess the pass ran on, `g(x)` is what the loop sent back.

`p history` reads the residual so far:

> falling, about ×0.039 per iteration

(mid-solve it reports the observed rate; at the end, the verdict on the solve).

Stop where it matters

```
difflow debug
difflow-dbg> break clear
all breakpoints cleared
difflow-dbg> break split
breakpoint before unit split
difflow-dbg> continue
continuing
── difflow-dbg ── @before_unit  split  iter 5  phase=solve
   ↳ unit split
   [recycle.F_n_hexane] recycle.F_n_hexane = 2.66667
difflow-dbg> p effluent
effluent:
  F_n_hexane         3.33333
  F_2_methylpentane 1.66667
  T                  340
  P                  101325
difflow-dbg> break clear
all breakpoints cleared
difflow-dbg> continue
continuing
── difflow-dbg ── @terminated  iter 6  phase=end  residual=2.874e-11  converged
   ↳ stop-at terminated
   [recycle.F_n_hexane] recycle.F_n_hexane = 2.66667
difflow-dbg> continue
continuing
difflow debug: solve finished, converged=True
```

`break split`	before unit `split`, every pass
`break 4`, `tbreak 4`	at iteration 4 (once)
`break if residual<1e-6 && iter>1`	a condition
`break on nonfinite`	where a NaN is born
`break on residual_rise`	the residual went up
`watch recycle.F_n_hexane`	print it at every stop

`break` lists them; `break clear` removes them.

STEP 4 · POST MORTEM

Where is the NaN born?

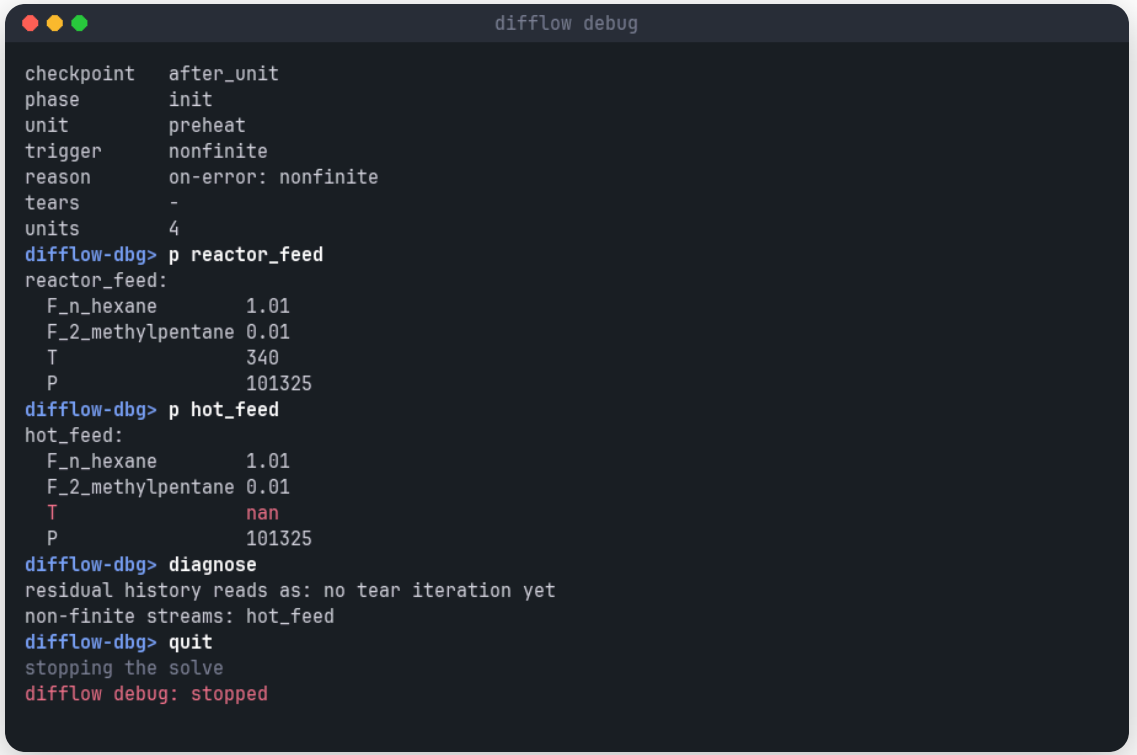
A preheater with a bug: fine above 400 K, NaN below.

```
def preheat(stream):
    T_out = stream["T"] + 10.0 * jnp.log(stream["T"] - 400.0)
    ...
```

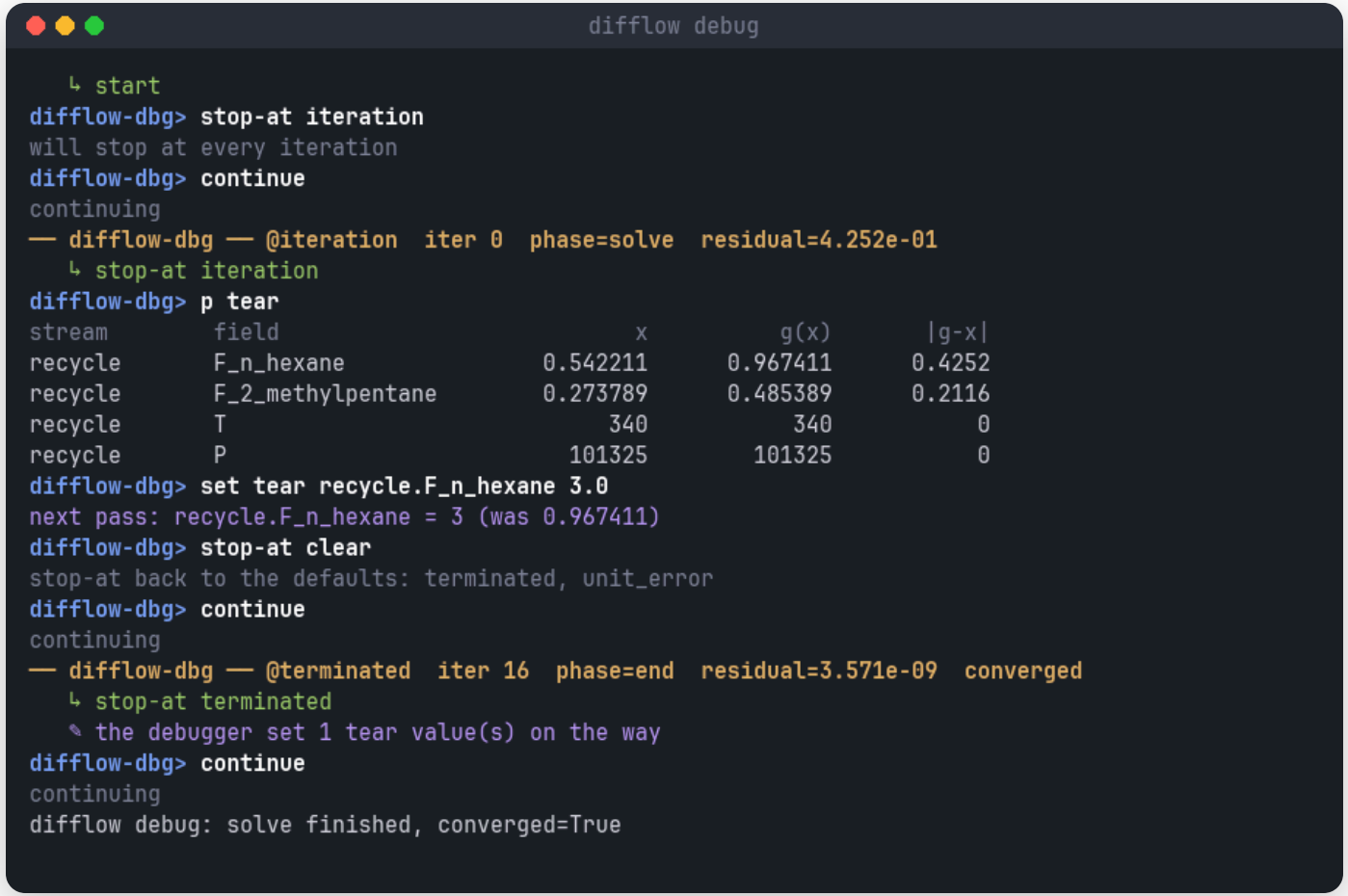
The plain solve says only `residual: nan`: by then every stream is NaN.

`--on-error` runs freely and stops **once**, at the first unit that turns *finite* inlets into a NaN (or raises, or the solve ends unconverged).

Here: `preheat`, in the **init** pass, before anything reached the tear.



`set tear STREAM.FIELD VALUE`



The one thing the debugger changes: the guess the **next pass** runs on.

- at an `iteration` pause it replaces that step
- at `after_accel` it edits the accelerated guess

An edit changes the **route**, not the fixed point: here 16 iterations instead of 6, the same product.

The edit is recorded in `fs.last_solve_debug_edits`, so an edited solve is never mistaken for a plain one.

FROM PYTHON

The same debugger in a script or notebook

```
from difflow import debug

with debug.attach():                  # a prompt here
    fs.solve(acceleration="wegstein")

with debug.attach(on_error=True):     # post mortem only
    fs.solve()
```

```
from difflow.debug.runner import DebugRunner

runner = DebugRunner(fs, {"acceleration": "wegstein"})
runner.start()                            # pause event
runner.command("break on nonfinite")      # result event
runner.command("continue")["next"]        # the next pause
runner.close()
```

`debug.attach()` gives the same `difflow-dbg>` prompt from a script, or in Jupyter as an input box under the cell.

`DebugRunner` runs the solve on a worker thread and hands back **one event per call**: the shape a server or an agent needs.

Detached, a solve is exactly what it was: one thread-local read per checkpoint. Attached, it only reads (apart from `set tear`), and `quit` raises `DebugQuit` rather than returning a partial answer.

STEP 6 · RECORD A SOLVE

`debug.record`: every checkpoint, kept

```
from difflow.debug.record import summarize, state_at

trace = debug.record(fs, {"acceleration": "wegstein"})
s = summarize(trace)
```

```
80 frames, 6 iterations, reads as: converged
frame  13  iter 0  4.25e-01  recycle.F_n_hexane
frame  20  iter 1  3.40e-01  recycle.F_n_hexane
frame  28  iter 2  6.52e-04  recycle.F_2_methylpentane
frame  36  iter 3  2.37e-04  recycle.F_n_hexane
frame  44  iter 4  2.24e-07  recycle.F_n_hexane
frame  52  iter 5  5.97e-08  recycle.F_n_hexane
frame  60  iter 6  2.87e-11  recycle.F_2_methylpentane
```

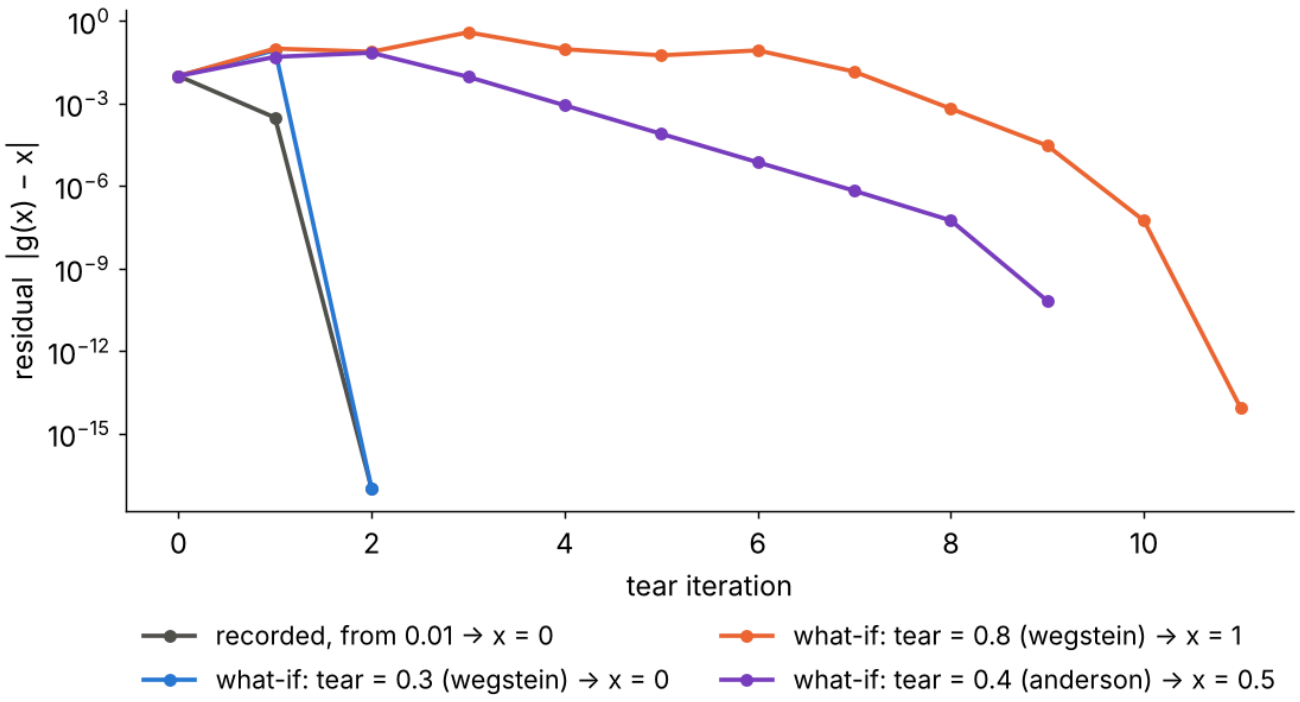
A recording solves once with nobody at the prompt.

Each frame holds only what that step wrote (the feeds, a unit's outlets, the tear iterate), so `state_at(trace, i)` is every stream at frame `i`.

`summarize` is the trace without its frames: one row per tear iteration, with the field the residual is mostly made of.

STEP 7 · WHAT IF?

Replay the solve with one value changed




```
replay = debug.record(fs, kw, baseline=trace["frames"],
    edits=[{"frame": k, "stream": "tear",
            "field": "F_A", "value": 0.8}])
compare(trace, replay)["same_answer"]   # False
```

A **replay**, not a fresh solve: everything before frame `k` is the recording (`diverged_at` checks it), and the accelerator keeps its history.

On a loop with steady states at 0 and 1: 0.3 comes back to 0; 0.8 reaches **1**. Anderson from 0.4 converges to **0.5**, the *unstable* middle state.

STEP 8 · DRIVE IT FROM A PROGRAM

`difflow debug --json`



The `difflow-dbg/1` protocol, one JSON object per line: a `hello` listing every checkpoint, event, command and capability, then a `pause` at each stop and a `result` per command. (Null fields left out here.)

The same debugger as MCP tools

`difflow mcp` serves:

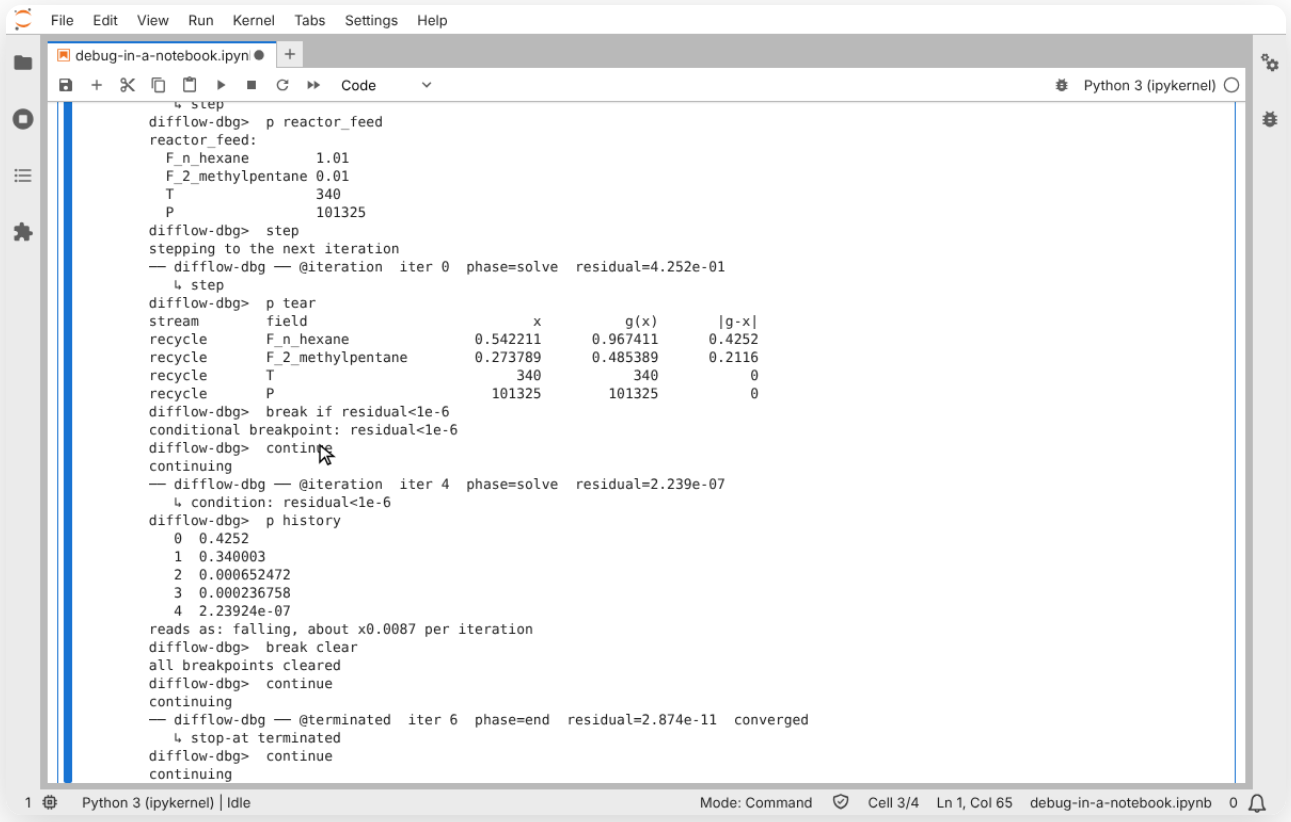
Tool	
`debug_guide`	the contract + the handshake
`debug_start`	pause before the first pass, or `on_error`
`debug_command`	any command; a resume returns the next stop
`debug_wait` / `debug_pause`	follow / stop a running solve
`debug_record`	a solve, summarized per iteration
`debug_trace_frame`	one frame in full
`debug_whatif`	a replay, compared: `same_answer`

The `fix_convergence` prompt uses them when the findings alone do not say why:

1. `debug_start`
2. `break on nonfinite`, `break on residual_rise`, `continue`
3. at the stop: `print tear`, `print history`, `diagnose`
4. `print` the unit at fault and its inlets
5. report the unit and the stream; `debug_stop`

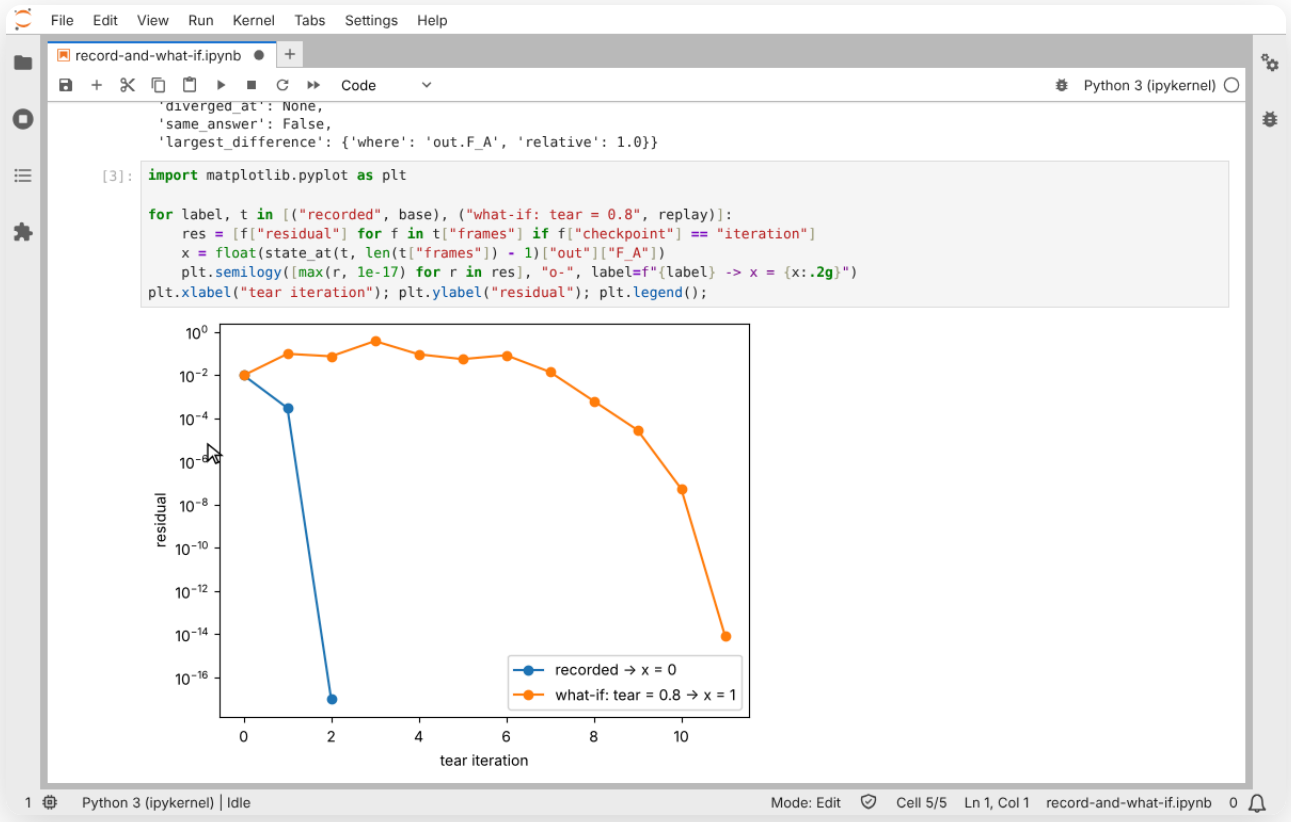
Each solve runs on a **copy**: debugging never changes the session's flowsheet.

The prompt, under the cell



JupyterLab, plain difflow: the cell runs `fs.solve` inside `with debug.attach():` and each `difflow-dbg>` prompt is an input box · `where` · `next` · `p reactor_feed` · `step` · `p tear` · `break if residual<1e-6` · `p history` · `continue`.

Keep a solve, then ask *what if*, as data



`debug.record(fs)` · `summarize` · a what-if at the first tear iteration with the tear set to 0.8 · `compare`: the same loop lands on the other steady state · plot both residual histories.

CHEAT SHEET

What to type

To	Type
start	`difflow debug plant.py`
post mortem	`difflow debug plant.py --on-error`
next unit / iteration / checkpoint	`next` · `step` · `stepi`
run on / to N / stop	`continue` · `run N` · `quit`
where am I	`info` · `where`
look	`p STREAM` · `p S.FIELD` · `p UNIT`
the tear	`p tear` · `p history` · `diagnose`

To	Type
break before a unit	`break UNIT`
break on a condition	`break if residual>1e3`
break on an event	`break on nonfinite`
watch a value	`watch S.FIELD`
try another guess	`set tear S.FIELD VALUE`
keep a solve	`debug.record(fs)`
ask what if	`record(edits=, baseline=)`

LIMITS, AND WHERE TO READ MORE

What it can and cannot step

- **Steps** the eager tear iteration: `acceleration="anderson"` (the default) and `"wegstein"`. `"none"` iterates inside optimistix: the debugger sees its start and end only.
- **Nothing fires under `jax.grad` or `jit`**: a traced solve is traced once, not run.
- **A unit is one step.** Its own inner solve (a CSTR's Newton iteration) is not stepped; what it reports in `info` is shown, and `break on inner_unconverged` stops on it.
- **It edits tear values only**, never parameters.

`docs/debugger.md` · `examples/69_flowsheet_debugger.ipynb` · `docs/agents.md` · `presentations/debugger/demo/*.ipynb`